QatarEnergy

قطر للطاقة

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IS PROCEDURE FOR THE CONTROL OF HOT WORK

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1.0 PURPOSE

The purpose of this Procedure is to address and outline controls and responsibilities associated with the safety of hot work operations carried out at QatarEnergy Idd El Shargi (QatarEnergy-IS) sites.

2.0 SCOPE AND APPLICABILITY

This Procedure describes the principal requirements for ensuring the safety of hot work (Naked Flame and Hot Work spark potential) operations on QatarEnergy-IS sites. Where contractors, operate their own PTW system these shall meet or exceed the requirements within this Procedure.

3.0 ROLES AND RESPONSIBILITIES

3.1 OFFSHORE INSTALLATION SUPERINTENDENT (OIS)

Authorize hot work activities which requires approval under IS-FOP-PRC-100 (Integrated Safe System of Work procedure – ISSoW).

3.2 AREA SUPERVISOR – (OS/OCS/WHMS/ODS-TRIOPS ONLY)

Countersign a hot work WCC to confirm his awareness in the ongoing work.

3.3 PERMIT APPLICANT/ WORKSITE SUPERVISOR

- In co-operation with Safety Officers / QatarEnergy-IS Representative and Permit Controller, complete the Hot Work Pre-Assessment (Refer Appendix A)
- Ensure the work area is clear of flammable materials, gases and vapors.
- Ensure adequate ventilation is available to remove vapors and fumes.
- Ensure spark containment enclosures and screens are properly erected and adequate for their intended use.
- Ensure that all safety precautions and conditions detailed in the WCC are adhered to and that all safety Procedures are fully implemented.
- Ensure all necessary safety and firefighting equipment is appropriately sited and readily available.
- Ensure a trained fire watcher is available for the duration of the hot work activities.
- Ensures that a Hot Work PTW/WCC is prominently displayed at the work area.
- Log worksite reading on the applicable PTW/WCC no less than every 60 minutes.

3.4 PERMIT CONTROLLER

- Ensure the Hot Work Pre-Assessment ((Refer Appendix A) has been completed by the Safety Officer /QatarEnergy-IS representative and Permit Applicant.
- Identify the Hot Work hazards and safety precautions necessary to ensure that hot work operations are performed safely.
- Review the hazard potential, precautions to be taken, and personnel protection specified whilst such operations are being performed.

- At the PTW/WCC planning review, shall identify potential work conflicts and ensure those activities shall be suspended or deferred.

3.5 SAFETY OFFICER / QATARENERGY -IS REPRESENTATIVE

- In co-operation with Permit Applicant and Permit Controller, complete the Hot Work Pre-Assessment (Attachment A).
- Inspect the spark enclosure/habitat to ensure it is fit for purpose prior to work commencing.
- advise on positioning of habitat and modification of existing escape routes as required.
- Ensures that a Hot Work PTW/WCC is prominently displayed at the work area.

3.6 FIREWATCHER(S)

Shall be trained in the responsibilities and duties of a fire watcher, Fire team member or fire team leader by an approved QatarEnergy-IS provider.

Fire watcher training shall include, but not limited to:

- Introduction of the fire triangle
- Responsibilities and duties of a fire watcher
- Safety precautions needed to prepare an area for hot work activity
- Types of fire and extinguishing equipment available
- Inspection of fire extinguishing equipment to ensure serviceability
- Emergency Procedures and communications in place during hot work.

The firewatcher shall always be present during hot work activities, this includes during break periods, or after the completion of hot work as fires can start from materials igniting after hot work is stopped. The work site shall be confirmed as safe before departure.

Consideration shall also be given to the need for a second firewatcher, for example in the area immediately below the work area where there is the potential for falling sparks or on the other side of bulkheads.

The firewatcher shall be responsible for continuous monitoring of the worksite and adjacent area during hot work operations to ensure the safety of personnel, plant and equipment.

Firewatcher responsibilities include the following:

- Take part in the ISSoW Risk Assessment
- Know where the closest ESD & MAC buttons are located
- Ensure appropriate fire extinguishers, charged fire water lines and gas monitors (where applicable) are available at the work area.
- Stopping hot work operations in the event that any hazardous or potentially hazardous situation is observed e.g. gas detection, sparks falling out of the spark containment enclosure.
- Understand the actions he has to take on the detection of gas at low and high levels
- Remaining at the work area until hot work has been completed or suspended and the automatic fire protection system has been reinstated.
- Communicate via Hand-held radio with CCR.

3.7 AUTHORISED GAS TESTER (AGT)

- Authorised gas testers (AGT's) are responsible for carrying out gas testing duties as per IS-FOP-PRC-100 and documenting gas readings on the PTW/WCC in accordance with this procedure.

Work Party shall:

- Undertake all hot work operations in a safe and effective manner as detailed in the Hot Work PTW/WCC.
- Ensure that they are fully aware of the hazards involved in the task and that they understand the conditions detailed in the PTW / WCC.
- Adhere to all applicable safety precautions prior to and throughout the duration of the work.

4.0 PROCEDURAL STEPS

4.1 GENERAL

Hot work naked flame activity at QatarEnergy-IS site locations should be avoided as far as reasonably practicable. To avoid hot work, a hot-work pre-assessment should be completed at the design and constructability stages.

Elimination of the requirement for hot work is the preferred option (such as pre-fabrication) and shall be considered at the design phase of each project or job.

Cold work substitution methods for site installation shall also be addressed, such internationally approved compression fittings and bolting systems in preference to hot work.

All work activities involving naked flame or spark potential hot work shall be undertaken in accordance with local Permit to Work (PTW) requirements. However, all work involving naked flame or spark potential hot work activities shall comply with the requirements detailed within this Procedure.

- PS1, Remote (WHM) locations and marine vessels (within the 500m zone) shall comply with IS-FOP-PRC-100 Integrated Safe System of Work (ISSOW). This shall be under the control of an appropriate Hot Work Control Certificate (WCC) and where necessary, a level 2 risk assessment shall be completed.

- QatarEnergy-IS facilities at Ras Laffan Industrial City (RLIC) and Halul Produced Water Handling Facility (PWHF) shall comply with IS-SCM-PRC-003 (Procedures for RLIC Safe System of Work) and Offshore Permit to Work Procedure (IP-OPS-012) at Halul PWHF.

- Drilling Rigs contracted to QatarEnergy-IS, shall comply under the contractor's PTW system and hot work Procedure for all work independent to their facility. Where Hot Work involves any QatarEnergy-IS facility, the default PTW shall be ISSoW.

The main hazards associated with hot work operations are:

- Fire or Explosion initiated by flame, sparks or hot material from the work site activities.

- Electric shocks from arc welding equipment.
- Weld bead particles or slag entering unprotected eyes during chipping.
- Inhalation of welding fumes.
- Noise, particularly from plasma arc welding, gouging operations and weld preparation.
- Radiation from UV and Infra-Red ('flash eye').
- Discomfort from heat source and uncomfortable postures.
- Additional heat load/stress from open flames and hot surfaces.

HOT WORK CATEGORIZATION

a. Hot Work – Naked Flame:

Naked Flame activities include:

- Work involving naked flames (flame cutting – burning, using Oxygen/Acetylene or
- Oxygen/Propane)
- Electrical & Diesel welding sets
- Use of portable grinders
- Brazing / soldering
- Grinding
- Operation of temporary diesel engines in hazardous areas
- Use of cartridge-operated fixing tools
- Use of a flare gun

b. Hot Work – Spark Potential

Spark Potential activities include:

- Use of high-temperature thermal calibrators
- Work involving explosives
- Use of equipment or work on pipework or vessels contaminated with pyrophoric scale
- In Hazardous Zones, the use of equipment or work involving:
- Needle gunning
- Inappropriately rated Ex equipment
- Opening live electrical junction boxes where terminals are exposed to the atmosphere
- Use of power tools including air / hydraulics in hazardous areas
- Use of low-power lasers
- Dry grit / shot blasting
- Spray painting (static spark)

4.2 KEY SAFETY MEASURES

- Ensure that prior to the PTW / WCC being issued for Hot Work Naked Flame, the habitat and worksite shall be checked by the facility Safety Officer or QatarEnergy-IS Representative to ensure it is correctly constructed.
- Ensure that all precautions and conditions detailed on the PTW / WCC are observed throughout the duration of the task.
- Ensure that no conflicting activity which could affect the safety or integrity of isolations/inhibits is allowed to take place.
- Regular checks shall be carried out by the work party throughout the work area when the PTW / WCC has been issued to ensure that conditions have not changed since time of authorization.
- Ensure gas tests are performed by an authorized gas tester (AGT) covering a 10 m radius of the worksite are carried out prior to commencement of the task, following periods of work suspension longer than 30 minutes (e.g. lunch breaks etc.) or after shift change. The readings shall be documented on the Hot Work Permit/WCC.
- The atmosphere shall be continuously monitored throughout the duration of the task using a portable area gas monitor. Readings shall be logged by the permit applicant on the applicable WCC no less than every 60 minutes.
- Prior to and throughout all Hot Work operations the conditions stipulated in the PTW / WCCs shall be complied with such as:
    - Restrictions on other operations in the work area
    - Isolations of fire and gas detection system
    - Presence of firewatcher
- Ensure that any area in which the fire detection/protection has been inhibited is continuously monitored by a fire watcher. Check that access passageways are clear and that all means of escape from the area are fully operational. Particular care shall be taken to ensure that hoses are run in such a manner that they do not present a trip hazard.
- Check that all flammable materials, including refuse, oil and grease, have been cleared away from the worksite and surrounding area prior to any hot work activity.
- Open hydrocarbon drains shall be plugged or covered during hot work activities.
- Check that fully operational fire extinguishers and charged fire hoses are in position at the worksite prior to any hot work activity.
- Check that ventilation systems and fume extraction equipment are operating satisfactorily.
- If the worksite is in a hydrocarbon process area, a survey with a gas camera shall be carried out prior to authorization of the PTW/WCC and continuous gas monitoring conducted by the work party.

- One trained fire watcher shall be available at the worksite during all hot work activities, should the welding sets be out of sight of the first fire watcher, then a second fire watcher shall be provided based on the ISSoW risk assessment.

4.3 HOT WORK SITE CONTROL

It is expected that no more than one live hot work naked flame permit should be active at any given time on live areas at PS1 complex. Similarly, it is expected that hot work naked flame will be limited to one live permit for remote areas (including Lift Boats and Drilling Rigs at remotes).

At the discretion of the OIS, further hot work naked flame activity may be considered only if fully justified and Risk assessment of cumulative risk is completed (e.g. ability to fully manage, resource and supervise the naked flame site and capability to provide full localised emergency response).

The number of permitted simultaneous hot work sites shall be strictly controlled and is governed by the resources available to safely monitor each such site. The review of resources shall take into account, but not be limited to, the available personnel for fire watch, gas monitoring, Risk Assessments and Safety Officer/personnel duties, together with their associated equipment and the amount of equipment in use at each site.

Hot Work sites shall be clearly marked, and distinct boundaries identified and established, with adequate detail noted on the relevant PTW and attachments.

4.4 HOT WORK - NAKED FLAME

Naked Flame hot work shall NOT be authorized if the proposed location is within Zone 1 unless the module or system affected by the HWNF work area has been fully depressurised, drained, flushed, isolated and N_2 purged and declared hydrocarbon free to create a safe zone.

Naked Flame hot work shall NOT be authorized if the proposed location is within or Zone 2 location or within a 10 ft (3 m) radius of an identified potential hydrocarbon leak source (piping or equipment) unless:

1. The module or system effected by the HMNF work area has been fully depressurised, drained, flushed, isolated and N_2 purged and declared hydrocarbon free to create a safe zone or,
2. The hot work location is fully enclosed and shielded within a pressurized habitat, and,
3. The work scope has been fully risk assessed and reviewed by the OIS.

4.5 PREPARATION

All hot work taking place in hazardous area shall be clearly defined to the OIS, by appropriate senior staff and by the Safety Officer in the form of a written work scope (in attachment to e PTW) complemented by detailed attachments, i.e.

- Reference number and date of the HW procedure
- Hot Work permit form and associated documentation such as Toolbox Talk or Confined Space Entry Certificate, when applicable tank plans

- Hazardous Area plan (where hot work takes place) to determine Area Classification
- Plot plan identifying location of work and distance to potential hydrocarbon leak sources, exact location of habitat (including layout control equipment and inlet and outlet air ducting)
- Review any existing deviations (ORA/SCRA) which could affect the worksite.
- Isolations and spading lists
- Job work pack.
- Task Risk Assessment;
- Fire Watcher duties
- Marked up P&IDs showing all relevant equipment, lines and isolations as per checklists
- Clear pictures of the plant undergoing hot work activities
- Drawings to illustrate the work scope clearly
- Any other document which can effectively complement the information on the hot work activity
- Conflicting operations to be identified and acted upon prior to commencement of Hot Work or activity (SIMOPs such as Bunkering and breaking containment to be considered)

4.6 HOSES

The following conditions apply with regard to hoses:

- Damaged portions of the hose are cut out and reconnected using the correct clamped connectors.
- A visually Inspection and leak test shall be carried out prior to use and any hoses found to fail the leak test or inspection shall be replaced
- Jubilee clips shall not be used to connect hoses.
- Hoses are correctly colour coded. Red - acetylene, Blue – oxygen and propane – Orange.
- Hose check valves and flashback arrestors are fitted immediately upstream of the welding torch or built into the welding torch, and flashback arrestors are fitted to both oxygen and fuel gas hoses immediately downstream of the regulator.
- Any hose in which a flashback has occurred will have its inner wall damaged and shall be discarded immediately.
- Eyes and ears are always protected when blowing gas to clear hoses and nozzles.
- When not in use and left unattended hoses including fire hoses shall be de-pressurized.
- At the end of a shift hoses and torches are withdrawn from the worksite and stored appropriately.
- Arrestors and regulators shall be used, inspected, and maintained in accordance with their manufacturer's instructions.

4.7 WELDING/ CUTTING TORCHES

The following conditions apply with regard to torch assembly:

- The correct torch is selected for the grade of work and the correct nozzle is used as per manufacturer's instructions.
- Regularly check all connections and equipment for faults and leaks. Use a proprietary leak detecting spray or solution suitable for use with oxy/fuel systems. Do not use soapy water or solutions containing grease or oils on oxygen systems
- Torches with leaks shall not be used.
- Torches are lit on acetylene first using a friction lighter and the oxygen valve used to adjust the flame.
- Gas from a torch shall not be used to blow clothing or equipment clean.

4.8 PROTECTIVE CLOTHING

Personnel conducting welding operations shall wear protective clothing as described in IS-HSE-PRC-005 Personal Protective Equipment Selection and Use.

4.9 METAL ARC WELDING

When metal arc welding is being performed the following additional points need to be considered:

- Insulated electrode holder
- Local isolation switch
- Fume extraction equipment
- Welding set transformer
- Work piece may require to be earthed
- Proper cable connections
- Welding leads shall be insulated, robustly constructed and big enough to carry the current safely
- Use of welding screens
- Use of an insulated box or hook to rest the electrode holder

4.10 SPARK CONTAINMENT SCREENS AND ENCLOSURES (HABITATS)

Process pipe work / equipment is considered to be 'live' if it:

- Contains any hydrocarbons at greater than atmospheric pressure.
- Is not positively isolated from a pressure source by means of spading or double block and bleed.
- Has not has its external power source positively disconnected, isolated and locked off where appropriate.

a. Application of Pressurised Habitats

If it is not possible to perform shutdown of hydrocarbon systems to perform HWNF (e.g. eliminate HWNF risk) then a suitably constructed positive pressurise habitat shall be utilised to carry out work within hazardous area zones:

- Pressurised habitats **NOT** authorised for use in Zone 0 or Zone 1 classified areas.
- Pressurised habitats are authorised for use in Zone 2 classified areas.

- Pressured habitats are considered good practice for use in non-hazardous zones in process areas / jackets.

b. Pressurised Habitat Construction

The pressurised habitat shall be constructed from suitable material (such as flame-retardant silicone coated fibre glass plastic tested in accordance with requirements of ANSI FM 4950-2007 Clause 5). A certified scaffold structure shall be used to provide the habitat framework.

The habitat workspace shall be sufficiently sized to comfortably house at least two persons. The habitat shall also be large enough to house any tools and equipment necessary for the work in hand. The interior shall be lined with fire blankets to reduced likelihood or any sparks or other incandescent material to make contact with the outer covering.

The habitat should be constructed to maintain an internal overpressure of 25-50 Pascal for a prolonged period. The exit shall be clearly marked, both inside and out. The exit shall be capable of being opened from either side and be of sufficient size to enable easy escape in an emergency. If the habitat obstructs a walkway, then suitable barriers shall be erected together with warning signs.

The habitat shall be illuminated by sufficient lighting, suitable for the zone in which the habitat is constructed, as advised by the Responsible Electrical Person.

c. Pressurization of Habitat

Pressurization shall be by air circulation drawn from a non-hazardous / safe zone. The exhaust vent shall be ducted from a point opposite to the inlet, to a non-hazardous open-air zone outside the habitat.

The habitat shall always remain closed and pressurised while hot work is in progress. It can only be opened after the hot work has ceased and the site made safe.

d. Ducting of Pressurised Habitat

Inlet and outlet ducts shall be protected and either marked with signs or guarded so that obstruction is prevented. The ducts shall be clearly marked, at both ends 'For Habitat Use-Do Not Remove".

e. Instrumented protection of pressurised habitat

<u>Zone 2 Applications</u>: As a minimum, a manometer shall be attached to the outside of the habitat and so placed that is can be read from the entrance. The instrument shall be suitable piped so that it can accurately read the pressure within. The habitat shall be tested to ensure that it can consistently hold overpressure of 25 Pascals.

All electrical power to the pressured habitat will be fed from existing platform EX rated socket outlets that are tripped by the fixed platform / facility shutdown system.

Portable gas monitors are required both inside and outside the habitat and shall also be placed at the air inlet ducting to ensure that air being used to pressurise the habitat is being drawn from a safe area.

Work can only commence when all monitors confirm that the atmosphere is free of flammable gas and the habitat is able obtain consistent positive overpressure of 25-50 pascals.

At the discretion of the OIS, Safety Advisor or work party, dependent upon perception of risk factors and outcome of the task risk assessment, in addition to the above minimum requirements, a self-contained, localised automated gas detection and habitat pressure shutdown system may be utilised. Upon detection of low habitat pressure (<25 Pascals), high temperature (40°C) or localised gas detection at air intake (5ppm H_2S or 5% Flammable gas) an automated shut down of the Hot work habitat site and initiation of localised alarm beacons will take place. Should it be required, the habitat can also be provided with an air lock door entry/ exit system to maintain pressure and EX rated air cooler at the inlet pipework to relieve heat stress.

f. Application of Screened Enclosures/ Screens

An enclosure constructed to contain ignition sources from contacting with flammable / combustible fluids or solids may be utilised for non-hazardous areas only, predominantly to protect people and equipment from harm (such as hot debris, sparks, flammable materials, welding arc light and avoid inadvertent trips of fixed F&G detection systems). As good practice a pressurised habitat should be considered the preferred option for all hot work naked flame activity.

All screen enclosures shall be built of materials that are flame-resistant e.g. scaffold with flame-resistant boarding and fire blankets.

- The enclosure shall be designed and built to contain sparks produced from the hot work task.
- The fire watcher shall monitor the enclosure for any spark discharge. If this hazard is realized, he shall stop the task until the repair to the enclosure is completed and spark containment is achieved.

5.0 REFERENCES TO OTHER DOCUMENTS

TABLE 1: REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal / External Document	Document Referencing
Integrated Safe System of Work Procedure (ISSOW)	IS-FOP-PRC-100		
RLIC Safe System of Work	IS-SCM-003		
IS WORK INSTRUCTION FOR SAFE TRANSPORTATION, HANDLING & STORAGE OF COMPRESSED GAS CYLINDERS	IS-FOP-WKI-020		
IS HSE TRAINING PROCEDURE	IS-HSE-PRC-002		
IS PROCEDURE FOR PERSONAL PROTECTIVE EQUIPMENT SELECTION AND USE	IS-HSE-PRC-005		
Offshore Permit to Work Procedure (QP)	IP-OPS-012		

6.0 DEFINITIONS

TABLE 2: KEY TERMS DEFINED

Term	Definition
Best Industry Practice	Where no regulation exists, best industry practice is the adoption of regulation and Procedure used within that specific industry in another place.
Pressurized Habitat	An enclosed area designed and constructed to contain maintain a positive pressure preventing the ingress of hydrocarbons or other hazardous gases whilst containing ignition sources within a safe

	environment (although normally used to create a safe zone within a hazardous area – they may be considered for use in non-hazardous areas to provide increased risk mitigation)
Hot Work	Work that can create an ignition source with sufficient calorific value to ignite flammable or combustible material which can be fluid, gas or solid e.g. Naked Flame, Welding, Grinding, Shrink wrap work for electrical terminations, Paint work holiday detection.
Method Statement	A statement from the licensed contractor detailing how the work is to be carried out safely
Spark Potential	Any work which under normal operation conditions does not produce an ignition source as described in “Hot Work” but has the potential to e.g. Blasting, Needle gun use, Sanding paintwork, Instrumentation and electrical testing & fault finding
Spark Containment Enclosure / Screen	A purpose-built enclosure to contain ignition sources from contacting with flammable / combustible fluids or solids
Shall	Mandatory action
Should	Recommended action
Zone 0	An area in which an explosive gas atmosphere is present continuously or for long periods (e.g. explosive atmosphere for more than 1000 hours per year)
Zone 1	An area in which an explosive gas atmosphere is likely to occur in normal operation (e.g. explosive atmosphere for more than 10, but less than 1000 hours per year).
Zone 2	An area in which an explosive gas atmosphere is not likely to occur in normal operation and, if it occurs, will only exist for a short time (e.g. explosive atmosphere present for up to 10 hours per year, but still sufficiently likely as to require controls over ignition sources).
Non-Hazardous Zone (Safe Area)	When the hazardous areas of a plant have been classified, the remainder will be defined as non-hazardous, sometimes referred to as 'safe areas'.

APPENDIX

APPENDIX A - HOT WORK PRE-ASSESSMENT

<table>
<tr><td colspan="8">Hot Work Pre-Assessment</td></tr>
<tr><td colspan="8">The Permit Applicant, Permit Controller and Safety Officer/QatarEnergy_IS Rep shall complete this assessment and use it to highlight potential hazards and controls required for the risk assessment. It shall be attached to the PTW/ WCC for review by the OIS</td></tr>
<tr><td colspan="2">W.C.C No.:</td><td colspan="2">Location:
Hazardous Zone:</td><td></td><td colspan="3">Date:</td></tr>
<tr><td colspan="8">Type of Hot Work</td></tr>
<tr><td colspan="2">Welding:</td><td>Cutting</td><td>Brazing</td><td>Soldering</td><td colspan="2">Grinding</td><td>Other (comment)</td></tr>
<tr><td>TIG</td><td>Stick</td><td></td><td></td><td></td><td colspan="2"></td><td></td></tr>
<tr><td colspan="4">Item</td><td>Yes</td><td>No</td><td>N/A</td><td>Comment</td></tr>
<tr><td colspan="4">Hot work requirement</td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="4">Can a cold work solution be found to complete the task? (If yes, STOP the job and re-assess)?</td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="4">Can the work piece be moved to a dedicated hot work fabrication shop? (If yes, please move work.)</td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="4">Gas Testing</td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="4">Are there any potential leak sources or combustible material within 35ft (10mters) of the work site?</td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="4">Has the area, jacket or module been surveyed by FLIR camera and confirmed no leaks present?</td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="4">Area Preparation</td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="4">Does the work area the including drains, contain any combustible/flammable material?</td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="4">Are there any additional precautions required for heat conducted/radiated through structures/partition walls etc.?</td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="4">Do Worksite barriers or the habitat impact on existing escape routes.</td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="4">Is there a requirement to highlight or signpost alternate escape routes?</td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="4">Isolations</td><td></td><td></td><td></td><td></td></tr>
</table>

Is there a requirement for fire and gas detection systems to be isolated.				
Does the construction of a habitat shall impact any fire or gas detection systems.				
Are other isolations required, e.g. machinery, etc?				
Emergency Response				
Do you have a job specific emergency response plan in place? Is there an adequate no of firewatchers for the specific task?				
HAZCOM				
Is there a requirement for specific PPE/ RPE, in addition to mandatory equipment, or engineering controls, due to the nature of the task? E.g. Welding galvanized metals.				
Other				
Are there any planned adjacent work activities with potential to create an explosive atmosphere through loss of containment or venting of hydrocarbons?				

Completed by		
Permit Applicant: Name/Sign:	Safety Officer/ QatarEnergy-IS Rep Name/Sign:	Permit Controller: Name/Sign:

APPENDIX B - FIRE WATCHER COMPETENCY CHECKLIST

Fire Watcher Competency Checklist				
The facilities HSE or QatarEnergy-IS representative shall interview the Fire Watcher(S), before the work activity commences to ensure knowledge of the roles and responsibilities of being a Fire Watcher.				
WWC:	Name:	Company:	Location:	Date:
Before Commencing the Work		**Yes / True**	**No / False**	**Comment**
Training / Qualifications				
Is the fire watcher trained in the duties and responsibilities of a fire watcher, fire team member or fire team leader by an approved QatarEnergy-IS provider? "If No – the fire watcher shall be replaced by a trained person"				
Safety Control / Equipment As Per WCC				
Is the fire watcher able to indicate where the nearest safety controls are located? • Nearest manual fire alarm point • Nearest fire water deluge activation • Nearest fire water hydrant • Nearest equipment power isolation switch (Welding) • Designated escape routes "Are they clear of obstructions?"				
Is the following safety equipment located at the worksite and is the fire watcher competent and able to demonstrate the correct operation of the following equipment? • Charged fire hoses • Correct type of portable fire extinguisher • Portable gas monitor • Portable radio / Radio check completed with CCR.				
Is the fire watcher able to describe what action, is required on activation of the portable gas monitor alarm or the platform GPA **"Stop the Job, Inform CCR & make the area safe"**				
Is the fire watcher aware that they shall not take part in any other work activity that could interfere with their duties as a fire watcher I.E Assist with the ongoing work activity, used to collect material or tools for the task?				
Is the fire watcher aware that he is responsible for monitoring the work and surrounding area for further hazards, and should a hazard present be committed to **"Stop Work"**				

Emergency Response			
Is the fire watcher able to describe the process for contacting the facility control room in an emergency by both “Radio and Telephone?			
Is the fire watcher aware of the QatarEnergy-IS ‘Stop Work” Authority?			

Completed by	
Fire watcher (name above) was found to be ☐ Non-Competent ☐ Competent to perform the duties and responsibilities of a fire watcher within QatarEnergy-IS under this ISSoW WCC. Personnel found non competent shall not be used as a fire watcher	
Safety Officer / QatarEnergy-IS Rep: Name/Signature:	Employee: Name/Signature:

APPENDIX C - PRESSURISED HABITAT CHECKLIST

PLATFORM		LOCATION		
COLD WORK PERMIT NO		DATE		
HOT WORK PERMIT NO		SHIFT	D Day Shift O Night Shift	

1. This checklist must be used to ensure that the habitat is fit for its intended purpose.
2. A new checklist must be completed every shift, and re-inspected regularly.
3. A copy of this checklist must be attached to the appropriate hot work permit.

Item	Description/ Checklist	Status Check		Comments/ Action List (if NO please state reason)
		Yes	No	
1	Has TRA and PTW been raised prior to commencement of work? Are all hot work equipment and accessories such as welding machines, cables, etc. of approved type to COMPANY?			
2	Is the scaffold valid for use and tagged? Is the scaffold of a suitable size and configuration to erect the habitat? (Ensure that the scaffold accommodates proper entry, exit, escape and airlock if required)			
3	Has the area been cleared of potential trip hazards and obstructions surrounding the access and emergency exits?			
4	Have all signs and barriers been installed to prevent unauthorized personnel from entering the work area?			
5	Is there a separate access and exit point? Are escape routes clearly marked and not obstructed? If applicable, has MOC been approved for escape route obstruction and has temporary escape route been identified, communicated to all parties and indicated by proper signage?			
6	If applicable, has the airlock been installed properly to provide a double door entry and exit arrangement?			
7	Is the habitat large enough to ensure that the panels will not be damaged by hot work activities? {It is recommended that hot work activities should be no closer than 50 cm to the panel. Should there be no alternative, a sheet metal screen may be used to prevent damage to the panels)			
8	Are any combustible materials inside the habitat, such as scaffold boards or ropes property protected with fire blankets and silica wall protectors in place to prevent welding slag damage?			
9	Is the air supply taken from a designated safe area? Is the inlet ducting securely attached ? Portabale Gas detection placed at air inlet and at work site. (Consider changes in wind direction)			
10	Is the outlet duct at the safe distance from any likely sources of flammable gases or liquids? Is the outlet ducting securely attached?			
11	Are the air supply and outlet ducts at least 5 meters apart?			
12	Has the manometer(s) been installed correctly and are visible?			
13	Do all instruments have valid calibration certificates not more than 6 months old?			
14	Are all electrical equipment connected securely using proper glanding and termination? (For pneumatic fans, the whip check and R-chip shall be in place before pressurizing)			

15	Are all electrical equipment and accessories certified as suitable for use under the prevailing hazardous area classification?			
16	Have all the zips and zip intersections been covered using flame retardant Velcro?			
17	Have all penetrations (piping, structural and adjacent plate faces) been properly sealed using sticky Velcro, clamps, cambuckles or other means?			
18	**Is the required overpressure (25 Pa) obtained and maintained?** Is there adequate ventilation rate within the habitat?			
19	Have any pinholes less than 10mm diameter been covered using the flame retardant tape?			
20	Is the entry tag placed on the outer door of the habitat? (The entry tag holder shall display RED sign when habitat is not fit for use and display GREEN sign which is dated when fit for use)			
21	Are the following hot work precautions in place? (Firewatchers, hoses, firewater pumps, extinguishers, etc.) Are these equipment fully functional and calibrated properly?			
22	Does the firewatcher have a good view of any areas where slag or spatter may fall? Is there a requirement for additional firewatchers outside and underneath the habitat? Does the habitat window allow for external monitoring?			
23	Is the habitat lighting sufficient and functioning properly? Is the lighting suitable for the prevalent hazardous area classification?			
24	Is there an adequate amount of spares available on site?			
25	If localised automated SD system been identified as additional control / mitigation requiremnet on the TRA/ Job pack review: Automatic shutdown devices been calibrated, tested and verified as working properly? Auto shutdown of Habitat on (5%LEL, 5ppm H2S at Inlet) and (Pressure < 25 Pascals) and (temperature > 40degC) tested? All visual beacons clearly visible and are audible alarms clearly heard within the habitat? Is the habitat provided with air lock entry / exit door arrangement? Has EX air cooller at air inlet been considered for heat stress?			
26	Are the work party trained and authorised for use of pressurised habitats			

HES Advisor	Name	
	Signature	
	Date/Time	
Construction Supervisor	Name	
	Signature	
	Date/Time	

APPENDIX D – REVISION HISTORY LOG

TABLE 1: REVISION HISTORY LOG

Revision Number: 01 **Date: 31/01/2021**

Item Revised	Revision Description	Page No.
Clause 3.1, 3.2 & 3.7	Additional information and requirements added to the clause 3.1, 3.2 and 3.7	-
Remarks: The document was updated based on feedback from OIS and specify the requirement on Pressurised Habitats.		

Revision Number: 02 **Date: 29/10/2023**

Item Revised	Revision Description	Page No.
All	Document simplified as part of QMR initiative.	All
Remarks:		

APPENDIX E – LIST OF ABBREVIATIONS

TABLE 4: LIST OF ABBREVIATIONS

Abbreviation	Description
AGT	Authorised Gas Tester
F&G	Fire and Gas
ESD	Emergency Shutdown
ISSoW	Integrated Safe System of Work
LEL	Lower Explosive Limit
HWNF	Hot Work Naked Flame
OIS	Offshore Installation Superintendent
ORA	Operational Risk Assessment

Abbreviation	Description
PTW	Permit to Work
PWHF	Produced Water Handling Facility
QatarEnergy -IS	QatarEnergy - Idd El Shargi
RLIC	Ras Laffan Industrial City
WCC	Work Control Certificate